

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460299.8906 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.021
Transit depth (Rp/Rs)^2	3.92 +/- 0.82 [%]
Orbital Inclination (inc)	81.7 +/- 1.15
Airmass coefficient 1 (a1)	1.004 +/- 0.019
Airmass coefficient 2 (a2)	0.0001 +/- 0.005
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	#3 - [350, 37]
Optimal Aperture	4.36
Optimal Annulus	5.43
Transit Duration (day)	0.0521 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.198 ± 0.021 vs 0.1594 (deviation 1.51σ)
Duration fit vs published	0.0521 d vs 0.0512 d (deviation 0.18σ)
Mid-time O-C	-5.0 min
Depth SNR	7.8
Residual scatter	1.89 %

Light curve

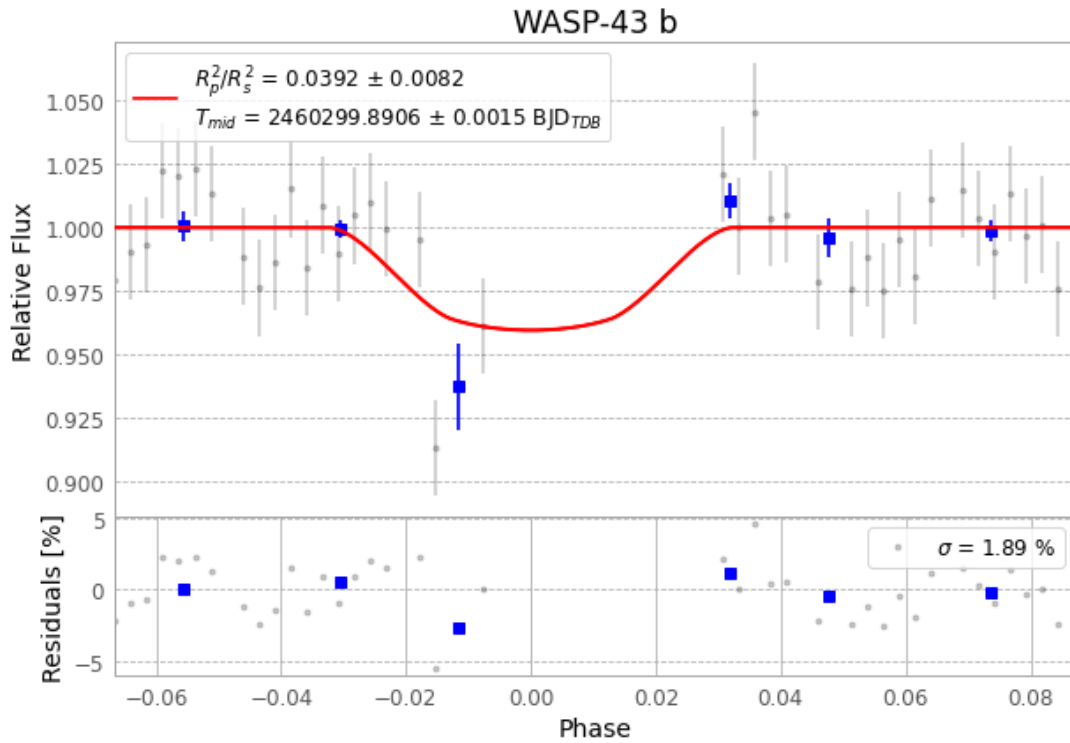


Figure 1 — FinalLightCurve_WASP-43 b_2023-12-21.png (SHA-256 93a768cca025b341...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [427, 232], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c89d54a848c4d50a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

53 FITS frames (35 MB), WASP-43231221080015.FITS → WASP-43231221110015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-231221.log (3a5607b9d519...), FinalLightCurve_WASP-43 b_2023-12-21.png (93a768cca025...), FinalLightCurve_WASP-43 b_2023-12-21.csv (a4ab6deb5c96...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 02:57Z.